IO-Link Communication Standard: Architecture, Smart-Device Functions

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***Abstract*—Modern automation systems require more than a simple on/off signal from a field device. Maintenance teams also need identification, parameter, status, and diagnostic information that conventional wiring cannot provide. This paper examines the IO-Link communication standard, standardized as the Single-drop Digital Communication Interface in IEC 61131-9. We explain how the physical connection, master-device structure, cyclic and acyclic data channels, and the IO Device Description (IODD) work together. We then evaluate IO-Link against the communication requirements of a small automated packaging and sorting cell. The comparison with conventional binary and 4–20 mA interfaces shows that IO-Link fits well at the last-meter connection of intelligent sensors and actuators. It improves transparency and simplifies device replacement, but does not replace controller-level Industrial Ethernet. Its main limitations are the 20 m cable restriction, modest bandwidth, master cost, and the need for higher-level security measures not included in the base protocol.**

***Index Terms*—IO-Link, IEC 61131-9, industrial communication, smart sensor, diagnostics, parameterization, Industry 4.0**

I. Introduction

Industrial machines depend on sensors and actuators that sit close to the physical process. A photoelectric sensor detects a bottle on a conveyor, a pressure sensor monitors a pneumatic line, and a valve changes the flow. For a long time these devices were wired directly to a programmable logic controller (PLC) using binary 24 V signals or 4–20 mA analog loops. This works well for simple cases, but it only brings the raw measurement into the controller. Information about device health, internal temperature, contamination, or configuration stays locked inside the device and is invisible to the system.

This becomes a real problem when production changes frequently. If a packaging line switches between bottle sizes, a technician must visit each sensor and adjust the threshold manually. When a sensor fails, the replacement must be configured from a datasheet. Both situations cost time and introduce human error.

IO-Link was developed to solve exactly this problem at the sensor and actuator level. IEC 61131-9 defines it as a point-to-point digital interface for exchanging process values, parameters, identity, and diagnostics with a smart field device [1]. Importantly, IO-Link is not a fieldbus. It does not compete with PROFINET, EtherCAT, or EtherNet/IP. Instead, it extends digital I/O at the last meter of the automation hierarchy [2], [3], as shown in Fig. 1.

We chose this topic because sensor-level communication is directly relevant to our coursework in industrial automation and real-time systems. The packaging cell example was selected because it combines binary detection, a measured value, and an actuator in a compact scenario that shows where IO-Link adds value and where simpler wiring is sufficient.

This paper answers three questions: (1) how does IO-Link communication work, (2) which smart-device functions provide practical value, and (3) under what conditions is IO-Link a better choice than conventional interfaces? The approach is standards-based and requirements-driven. We do not report measured latency values because no physical IO-Link test bench was available during this study.

II. Background

Industrial control networks have stricter requirements than office networks. Determinism, availability, and robustness matter more than peak throughput, because losing communication can stop a physical process [4], [5]. More recent automation architectures add Industrial Ethernet and cloud connectivity, creating a vertical data path from the sensor to enterprise systems [6].

Field connectivity can be thought of in three levels. Conventional I/O connects each signal directly: simple but carrying almost no device information. Fieldbus and Industrial Ethernet enable richer data exchange between distributed controllers, but a gap often remains between an intelligent sensor and the nearest I/O module. IO-Link fills this gap by adding a digital communication layer to the familiar three-wire connection.

Two properties of modern sensors make this worthwhile. Even low-cost devices now contain microcontrollers that can report internal temperature, operating hours, contamination, and signal quality alongside the main measurement. IO-Link provides one standardized interface for accessing this data across manufacturers.

The base protocol is a local, wired, point-to-point link with no Ethernet addressing, no multi-node routing, and no built-in encryption or access control [7]. Security must be handled at the system level through physical protection, the master, the PLC, and the engineering network.

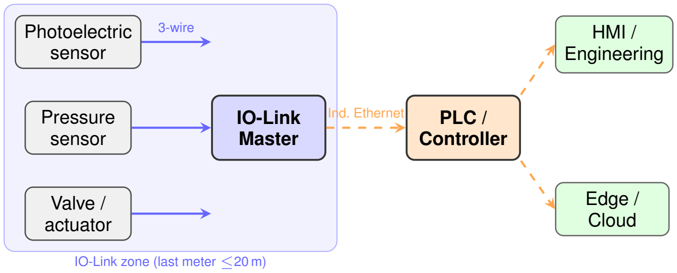

Fig. 1. Position of IO-Link in an automation system. Each port is a dedicated point-to-point link; the master bridges field devices to the Industrial Ethernet controller network.

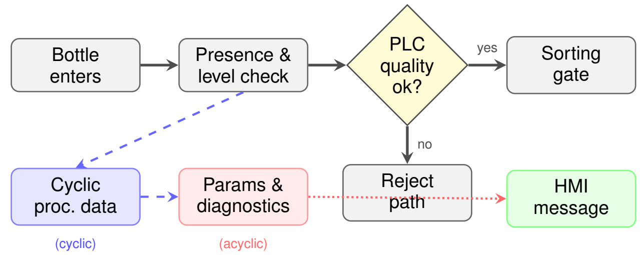

Fig. 2. Information flow in the packaging cell. Solid arrows show the physical process; dashed and dotted arrows show the IO-Link cyclic and acyclic data channels respectively.

III. IO-Link System Architecture

A. Components and Physical Interface

An IO-Link system consists of a master and one device per port. Devices can be sensors, actuators, valve terminals, or I/O hubs. The master controls all communication and maps device data into the higher-level control network, typically over PROFINET, EtherNet/IP, or EtherCAT.

Figure 1 shows how IO-Link sits in the automation hierarchy. Each master port makes a dedicated point-to-point link to one field device. The master then connects upward to the PLC over Industrial Ethernet.

The physical connection uses three wires: L+ and L− for power, and C/Q for the signal. The C/Q line carries either a conventional switching signal in SIO mode or serial IO-Link communication. Standard M8 or M12 connectors support unshielded cables up to 20 m. Port Class A covers most sensors; Port Class B adds a separate auxiliary supply for actuators with higher power demand [3].

The 20 m cable limit is the most important placement constraint. In a compact machine this is rarely a problem, but in larger plants masters must be distributed. A useful feature is backward compatibility: a master port can be configured as IO-Link, digital input, or digital output, enabling gradual migration from conventional wiring.

B. Communication and Data Types

The master always initiates communication; devices only respond. Three data rates are defined: COM1 at 4.8 kbit/s, COM2 at 38.4 kbit/s, and COM3 at 230.4 kbit/s. Each device supports one rate; a conforming master detects it automatically.

The protocol separates data into four types:

- **Process data** — exchanged cyclically, up to 32 octets per direction per cycle.
- **Value status** — indicates whether the current cyclic data are valid.
- **Device data** — read or written acyclically by index and subindex; covers identity, parameters, and commands.
- **Events** — report errors and maintenance conditions such as wire break, contamination, or overtemperature.

The cyclic and acyclic channels operate in parallel: a PLC receives measurements every cycle while an engineering tool reads diagnostics in the background without interruption. Error detection uses a checksum on every frame; if a frame fails, the master retries up to twice before reporting a fault [3].

End-to-end latency depends on the device minimum cycle time, data length, communication rate, master processing, fieldbus update, and PLC scan. These must be summed and verified for each installation.

C. IODD, Parameterization, and Diagnostics

Every conforming IO-Link device ships with an IO Device Description (IODD) file [8]: a machine-readable XML file describing communication properties, identity, process-data structure, parameters, and diagnostic events. Engineering software imports the IODD and presents device settings in a consistent interface across manufacturers.

Parameter data storage lets the master save all device parameters and restore them automatically when a compatible replacement is connected. The diagnostic channel provides specific fault information—contaminated optics, overtemperature, low supply voltage—rather than a generic fault bit. These diagnostics only reduce downtime when the control program and maintenance team actively use them.

IV. Application Scenario and Evaluation

A. Automated Packaging and Sorting Cell

The evaluation scenario is a small packaging cell in which bottles move on a conveyor, are checked, and are directed to an accepted or rejected path. The devices are: a photoelectric sensor for bottle detection, a distance sensor to check fill level, a pressure sensor for the pneumatic supply, and a valve actuator for the sorting gate. All connect to an eight-port IO-Link master over PROFINET to the PLC, with an HMI for process display and diagnostics.

Figure 2 shows the information flow. Cyclic process data travel from sensors to the PLC continuously. The PLC makes the sorting decision and drives the gate. Recipe parameters and diagnostics flow through the acyclic channel to and from the HMI.

B. Communication Requirements

Table I defines the requirements used for evaluation. The 10 ms update target is specific to this conveyor application.

TABLE I
REQUIREMENTS FOR THE PACKAGING CELL

ID	Requirement	Evaluation criterion
R1	Sensor values reach PLC within a 10 ms application update target.	Device cycle, master, field-bus, and PLC scan checked as a chain.
R2	Product thresholds changed from HMI without visiting each sensor.	Indexed parameter write via IODD engineering tool.
R3	Device faults show a meaningful cause on the HMI.	Value status, events, and device diagnostics.
R4	Replacement sensor recovers approved settings automatically.	Identity check and master data storage.
R5	Standard 24 V wiring within 20 m.	Three-wire connection and cable limit.
R6	Interface integrates into existing PROFINET system.	Master as fieldbus-independent gateway.
R7	Unauthorized parameter changes controlled.	Physical zone and PLC/master access control.

C. Evaluation Method

The evaluation is qualitative and requirements-driven. Each requirement is mapped to protocol functions from the standard, noting where IO-Link satisfies it fully, partially, or requires additional system-level measures. Conventional 24 V digital I/O and 4–20 mA analog are used as baselines. No measured latency is reported because no physical test bench was available.

V. EVALUATION AND DISCUSSION

A. Performance and Requirement Fit

R1 – Timing. IO-Link provides master-controlled cyclic communication, but meeting the 10 ms target is not automatic. The full end-to-end timing chain must be summed:

$$t_{\text{total}} \approx t_{\text{dev}} + t_{\text{master}} + t_{\text{fieldbus}} + t_{\text{PLC}} \tag{1}$$

As a rough estimate for this cell: a COM2 device with a 2 ms minimum cycle, 2 ms master processing, 4 ms PROFINET update, and 2 ms PLC scan gives approximately 10 ms end-to-end — exactly at the target with no margin. A COM1 device or a longer PLC scan would exceed it. The designer must verify each element during commissioning.

Raw bandwidth is intentionally modest. COM3 at 230.4 kbit/s is far below Industrial Ethernet, but sensor messages are small. Up to 32 octets per cycle per device is sufficient for measurements, status bits, and actuator commands.

R2, R3, R4 – Smart device functions. These three requirements are where IO-Link provides the clearest benefit. Remote parameter access via the IODD allows an HMI or PLC to load a product recipe without visiting the sensor. Diagnostic events identify the specific fault cause rather than a generic error bit. Parameter data storage means a replacement device is configured automatically. Together these functions reduce changeover time, shorten mean time to repair, and lower configuration error risk.

TABLE II
COMPARISON OF FIELD-DEVICE INTERFACES

Criterion	Digital I/O	4–20 mA	IO-Link
Information	One on/off state	One process value	Values, status, params, events
Direction	One-way	Device to ctrl	Bidirectional
Parameters	Local only	Local only	Remote via IODD
Diagnostics	Wire state	Limited	Standardized events
Replacement	Manual	Manual	Auto restore
Cable	24 V std	Dedicated line	3-wire, ≤20 m
Cost	Lowest	Low	Master adds cost
Best fit	Switches	Long-distance	Smart sensors, flex. machines

R5, R6 – Wiring and integration. The three-wire 24 V connection uses familiar field wiring. PROFINET integration is handled by the master as a transparent gateway. The 20 m limit suits a compact cell; in a larger plant, masters must be distributed closer to devices.

R7 – Security. Only partly covered by the base protocol. The point-to-point topology limits network exposure, but the protocol provides no encryption, authentication, or access control [7]. Parameter changes must be protected through PLC user management, master access control, and a secured engineering network.

B. Comparison with Conventional Interfaces

Table II summarizes the main trade-offs. Conventional solutions remain appropriate in many cases. A digital input is simpler and cheaper for a fixed limit switch. A 4–20 mA loop handles long distances and resists electrical noise. IO-Link is the better choice when additional device information saves real time during commissioning, changeovers, or fault diagnosis.

C. Advantages and Limitations

The main advantages are: richer field data without extra cables; vendor-independent device descriptions; remote configuration; automatic parameter backup; detailed diagnostics; backward-compatible ports; and fieldbus-independent integration. These are most valuable in flexible machines with frequent product changes or condition monitoring needs.

The limitations are equally real. Each device occupies one master port. The 20 m cable limits placement options. Bandwidth is designed for sensors, not cameras or large data streams. The base protocol has no safety or security functions. Most importantly: diagnostics that nobody monitors provide no value. The benefits of IO-Link depend on the surrounding software, processes, and people.

VI. PRACTICAL INTEGRATION NOTES

Based on the standard documentation and our evaluation, several points are relevant for an engineer planning a similar installation.

TABLE III
PROPOSED COMMISSIONING CHECKS (NOT EXPERIMENTALLY VERIFIED)

T	Procedure	Expected result
T1	Timestamp sensor change at PLC under load.	Worst-case update meets 10 ms target.
T2	Load product recipe; read parameters back.	Values match recipe; invalid write gives clear fault.
T3	Replace sensor with compatible spare.	Master restores settings without local adjustment.
T4	Disconnect cable; trigger device warning.	HMI distinguishes comm. loss from device fault.
T5	Attempt parameter change as unauthorized user.	Change blocked; physical access restricted.

Planning. Obtain the IODD for every device and verify communication rate, cycle time, process-data length, and diagnostic events before ordering. Select Port Class A for sensors and Class B for actuators needing separate power. Place the master within 20 m of all its devices.

PLC programming. Map cyclic data into clearly named structures. Always check the value status bit before trusting a measurement. Keep parameter and diagnostic access in the acyclic channel, separate from the fast control loop. Translate device event codes into plain-language HMI messages.

Parameter management. Separate recipe parameters from protected calibration or service parameters. Read back a parameter after writing to confirm success. Configure master data storage deliberately so it backs up an approved configuration, not an accidental change.

Security. Keep field cables inside a physically controlled zone. Use individual accounts and role-based permissions for engineering access. Route remote access through the plant security architecture.

Table III proposes commissioning checks for the cell. These are not results we measured; they are tests to be performed during commissioning to confirm the requirements in Table I are met.

VII. SELECTION GUIDANCE AND LIMITATIONS

IO-Link should be chosen based on whether its functions solve a real problem, not because it is newer technology. A fixed binary sensor installed once and rarely changed does not justify a master port. In contrast, an adjustable distance sensor used across multiple product variants benefits directly from remote thresholds, contamination warnings, and automatic parameter recovery.

A practical selection follows three steps: (1) check the control requirement: signal type, update deadline, safety relevance, and cable distance; (2) consider lifecycle needs: product changes, diagnostics, replacement frequency, and maintenance skill; (3) check system constraints: master ports, fieldbus compatibility, IODD availability, and engineering tool support. Table IV summarizes the key decision questions.

TABLE IV
TECHNOLOGY SELECTION QUESTIONS

Question	Design consequence
Only one fixed on/off state?	Prefer digital I/O unless diagnostics add clear value.
Long-distance continuous signal?	Consider 4–20 mA; check 20 m IO-Link limit.
Thresholds change regularly?	IO-Link parameterization reduces manual setup.
Detailed fault info needed?	Verify IODD includes the required events.
Fast replacement by general staff?	Use identity checking and tested data-storage settings.
Safety-critical or remotely exposed?	Apply full safety and security architecture above IO-Link.

This study has several limitations. The evaluation is based on standard documents and a representative example, not laboratory measurements. The timing estimate in (1) uses typical values for illustration; actual values depend on the specific devices, master, fieldbus, and PLC task configuration. The paper covers only wired base IO-Link; IO-Link Safety, IO-Link Wireless, and cloud integration each require separate study.

VIII. CONCLUSION

This paper examined wired IO-Link as the last-meter interface between intelligent sensors and actuators and an industrial PLC. The standard combines a 24 V point-to-point connection with cyclic process data, acyclic parameter access, standardized device descriptions, diagnostic events, and parameter backup. For the packaging cell, these functions address the main requirements: predictable sensor updates, remote recipe change, meaningful diagnostics, automatic device replacement, and PROFINET integration.

IO-Link is not always the right choice. Conventional digital or analog I/O is simpler, cheaper, and sufficient for signals that never need remote configuration or detailed diagnostics. The strongest case appears when repeated changeovers, troubleshooting, or device replacement would otherwise cause production downtime. Timing, cable distance, security, and cost constraints must all be part of the design decision.

Future work could implement the described cell and measure actual end-to-end update time, retry behavior, and replacement time with a specific master and device. A comparison with IO-Link Wireless would also be valuable for applications where cable routing to each sensor is impractical.

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